Printed Pages– 10 Roll No. .....

341151(41)

B. Pharmacy (First Semester) Examination, Nov.-Dec. 2021

(PCI Scheme)

(Pharmacy Branch)

HUMAN ANATOMY and PHYSIOLOGY-I

(Theory : BP101T)

Time Allowed : Three hours

Maximum Marks : 75

Note : Question paper is of three parts i.e. (A), (B) and (C). Part A consists of 20 MCQs each of 1 mark. All questions are compulsory. Part B consists of 3 long answer questions of which attempt any two. Each of 10 marks. Part C consists of 9 short answer questions, attempt any seven questions. Each of 5 marks. Make a clear diagram wherever necessary.

Section-A

(Objective Type Questions) 20×1=20

Note : Attempt all questions, each carries 1 mark.

1. Multiple Choice Questions (MCQs)

(i) The cell membrane is :

(a) Solid

(b) Semi solid

(c) Fluid

(d) Gaseous

(ii) Protein synthesis takes place in :

(a) Ribosome

(b) Endoplasmic reticulum

(c) Nucleus

(d) Golgi complex

(iii) Cell power house is :

(a) Mitochondria

(b) Golgi apparatus

(c) Lysosomes

(d) Endoplasmic reticulum

(iv) Sodium-potassium pump is an example of :

(a) Active transport

(b) Passive transport

(c) Facilitated diffusion

(d) Osmosis

(v) The total number of bones in adult human body is :

(a) 206

(b) 210

(c) 260

(d) 236

(vi) The transformation of cartilage into bone is called :

(a) Osteoporosis

(b) Haemopoesis

(c) Ossification

(d) Haemolysis

(vii) Patella is a bone found in :

(a) Carpus

(b) Ankle

(c) [illegible]

(d) Tendons

(viii) The vertebrae are separated from each other by

(a) Blood vessels

(b) Nerves

(c) Cartilaginous

(d) Muscle

(ix) The brain stem comprises of :

(a) Medulla, Pons and midbrain

(b) Thalamus, hypothalamus and pituitary complex

(c) Tegmentum and tectum

(d) Medulla, pons and cerebellum

(x) ANS regulates the activity of all except :

(a) Glands of GIT

(b) Sweat

(c) Heart

(d) Skeletal muscle

(xi) In normal adult volume of CSF is approximately :

(a) 250 ml

(b) 350 ml

(c) 150 ml

(d) 50 ml

(xii) In spinal cord, grey matter :

(a) Is surrounded by white matter

(b) Surrounds the white matter

(c) Does not exist

(d) Forms nerves

(xiii) Which of the following is structural unit of nervous system :

(a) Alveoli

(b) Nephron

(c) Neuron

(d) Leucocyte

(xiv) Which of the following [illegible] from the brain or spinal cord [illegible]

(a) Mixed nerves

(b) Motor nerves

(c) Sensory nerves

(d) All of the above

(xv) Which of the following is divided into parasympathetic nervous system and sympathetic nervous system :

(a) Central Nervous System

(b) Peripheral Nervous System

(c) Autonomic Nervous System

(d) Spinal cord

(xvi) Which of the following glands are present in eyes :

(a) Pituitary gland

(b) Lacrimal gland

(c) Thyroid gland

(d) Mucus gland

(xvii) Which of the following layer of skin does not contain blood vessels :

(a) Cutis

(b) Dermis

(c) Corium

(d) Epidermis

(xviii) Posterior pituitary releases which of the following hormones :

(a) TSH and GH

(b) TSH and LH

(c) ADH and prolactin

(d) Oxytocin and Vasopressin

(xix) Largest endocrine gland is :

(a) Pituitary

(b) Thyroid

(c) Adrenals

(d) Pancreas

(xx) Pancreatic β-cell secretes :

(a) Pepsin

(b) Insulin

(c) Glucagon

(d) Somatostatin

Section-B

(Long Answer Type Questions) 2×10=20

Note : Attempt any two questions out of 3 questions. Each question carries 10 marks.

2. Define cell with well labelled diagram of different orgenells. Write the functions of cell and its compositions.

3. What are the various classifications of nervous system? Write in details about structure and functions of sympathetic and parasympathetic nervous system.

4. (a) Write the locations, structure and functions of Adrenal gland.

   (b) Write the locations, structure and functions of pancreas.

Section-C

(Short Answer Type Questions) 7×5=35

Note : Attempt any seven questions out of nine questions. Each question carries 5 marks.

5. Discuss homeostasis and importance of negative and positive feedback mechanism.

6. Write a detail note on nervous tissue.

7. Write classification of joints and their functions.

8. Discuss about structure and functions of skin.

9. Write a detail note on neuromuscular junction.

10. Describe reflex activity with well labelled diagram.

11. Explain physiology of hearing with neat and clean diagram.

12. Write the functions of posterior pituitary hormones

13. What are cranial nerves? Write in details about their functions.